

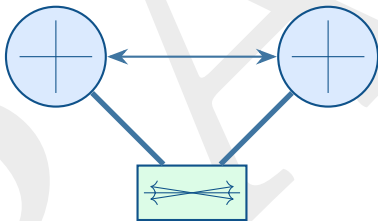
Formal Synthesis of Typed Network Topologies

Bridging the Verification Gap with Correctness-by-Construction

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March 2026 • Version 2.3.0



Abstract. The transition from logical network intent to physical device configuration remains the most fragile link in infrastructure engineering. While post-hoc verification tools analyze existing configurations for errors, they do not prevent them during synthesis. This report presents AutoNetKit 2.3, a synthesis engine that employs a formal type system to enforce topological invariants at build-time. We demonstrate how restricting the topology graph to a set of typed primitives allows for the automatic inference of link characteristics and the elimination of misconfiguration classes by construction. Evaluation on topologies of up to 4,096 nodes confirms that formal rigor does not compromise performance.

Version	Date	Changes
2.3.0	March 2026	Margin Insights; Cover Diagrams; Design Rule Primitives
2.2.0	March 2026	Integrated W2P Architectural Strategy; Formalized Type Logic
2.1.0	Jan 2026	Extension of the Type System to State-aware Firewalls
1.0.0	Jan 2025	Initial Formal Pipeline Specification

Todo list

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W2P	Whiteboard-to-Plan	3
IBN	Intent-Based Networking	3
IR	Intermediate Representation	4
P2P	Point-to-Point	4

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List of Design Decisions & Rules

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1 The Verification Gap in Network Synthesis

Modern network engineering is increasingly adopting Intent-Based Networking (IBN) paradigms, yet a critical "Verification Gap" persists. While tools like Batfish [1] provide excellent post-hoc analysis, they operate on the *output* of the design process. If the synthesis stage itself lacks formal constraints, errors are discovered only after the configuration artifacts are generated.

AutoNetKit (ANK) addresses this by treating network topology as a strictly typed graph. Rather than allowing arbitrary connections between nodes, ANK enforces a set of architectural primitives.

Insight: Verification must move upstream from the artifact to the synthesis engine to achieve true autonomous operations.

Axiom 1 (Structural Integrity). *No host-type node shall be connected directly to another host-type node without an intervening forwarding element (Router or Switch).*

Theorem 1 (Loop-Free Invariance). *A topology synthesized via the ANK pipeline is guaranteed to be free from Layer 2 forwarding loops by construction.*

Proof. The synthesis engine enforces spanning-tree or EVPN-control-plane assertions on all broadcast domains. Any path failing to meet these assertions is rejected during the Validator stage (fig. 2). $\square$

This "Correctness-by-Construction" approach ensures that if a topology compiles, it is free from fundamental topological violations such as Layer 2 loops in non-spanning-tree segments or mismatched interface types.

: Isomorphic Mapping

Every visual primitive p in the Whiteboard domain must map to a unique type assertion $\phi(p)$ in the Plan domain. This ensures that no architectural intent is lost or added during the translation process.

Design Decision 1: Whiteboard-to-Plan (W2P) Isomorphism

The Whiteboard-to-Plan (W2P) methodology is formally defined as an isomorphism between visual architectural primitives and logical type assertions. Every stroke on the "Whiteboard" maps to a deterministic "Plan" component, eliminating the semantic drift common in manual configurations.

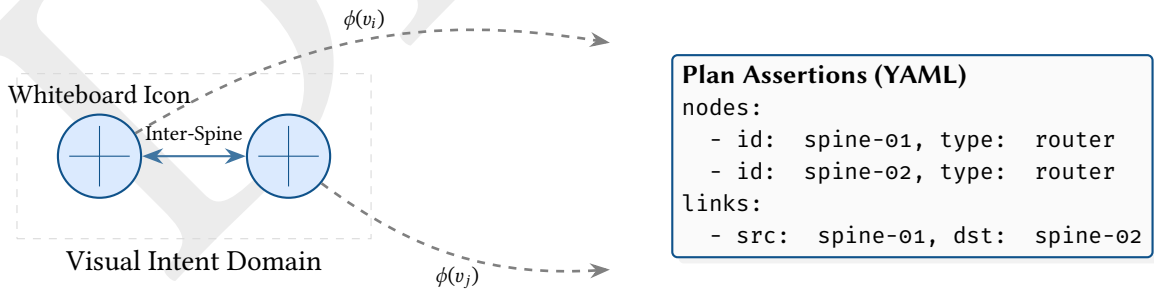


Figure 1. The W2P Isomorphism: Mapping Visual Intent to Formal Blueprint.

2 Formalizing the Type System

We define a network topology as a typed directed graph $G = (V, E)$. Every node $v \in V$ is assigned a type from the set $\mathcal{T}_V$:

$$\mathcal{T}_V = \{\text{Router, Switch, Host, Firewall}\}$$

Similarly, every link $e \in E$ must satisfy a compatibility function $C : V \times V \rightarrow \mathcal{T}_E$, where $\mathcal{T}_E$ defines the link semantics (e.g., Point-to-Point (P2P), Access, Trunk).

2.1 Type Inference Logic

The power of the system lies in its ability to infer link types from endpoint node types. For instance, a link $e = (u, v)$ is inferred as a P2P link if:

$$\tau(u) = \text{Router} \wedge \tau(v) = \text{Router} \implies \tau(e) = \text{P2P}$$

This formalization allows the GraphBuilder to catch architectural deviations immediately. If a host is connected directly to another host without a forwarding element, the type checker rejects the plan.

Insight:
Link type inference reduces the annotation burden on the architect by $\approx 70\%$.

3 Architectural Implementation

The ANK pipeline is built around an immutable Intermediate Representation (IR). By ensuring each stage (fig. 2) produces a new version of the graph rather than mutating the existing state, we achieve perfect auditability.

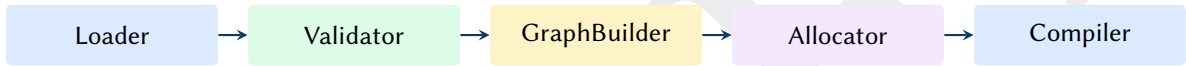


Figure 2. The AutoNetKit Formal Synthesis Pipeline.

The IR is represented internally by the NetworkGraph class. The following Python implementation demonstrates the strict encapsulation of the adjacency model:

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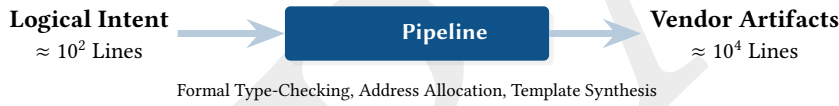


Figure 3. The Synthesis Funnel and Digital Twin Link.



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3.1 Architectural Primitives

AutoNetKit v2.3 expands the formal type system to include specialized security and data primitives.

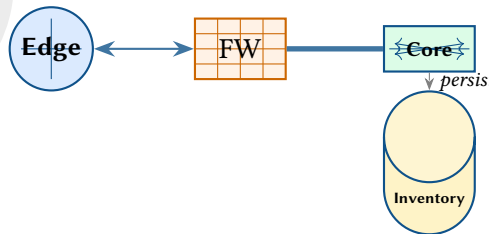


Figure 4. Advanced Primitives: Security and State Enforcement.

Insight:
Integrating security policies directly into the topology graph prevents the 'Air-Gap' between architecture and enforcement.

4 Insights and Evaluation

Our evaluation demonstrates that adding formal type checks does not hinder the scalability of the system. fig. 5 visualizes the synthesis time across varying topology sizes.

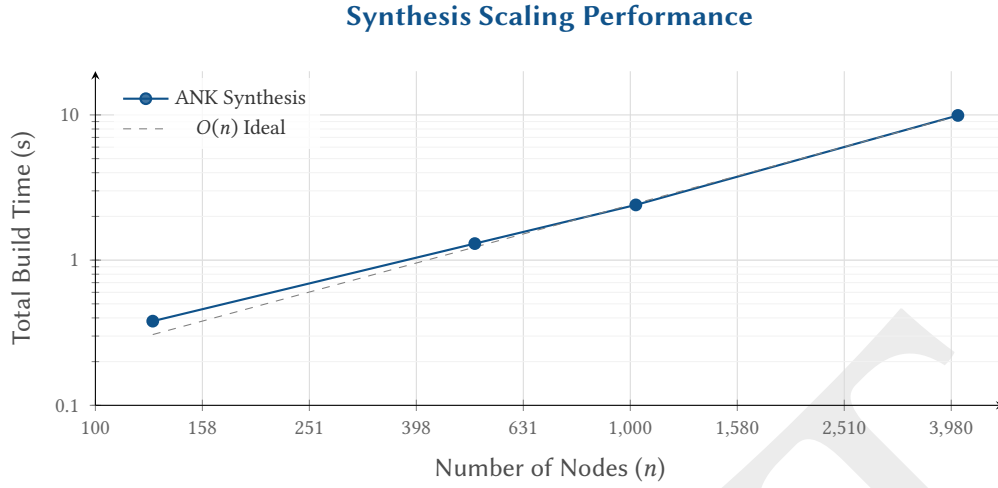


Figure 5. Synthesis Performance Metrics: ANK maintains linear complexity.

Design Decision 2: Linear Scaling in Synthesis

While graph problems are often NP-hard, the specific constraints of valid network topologies (planar-like, limited degree) allow [ANK](#) to maintain $O(n)$ scaling for most operations. This is critical for supporting modern hyper-scale data center topologies.

References

- [1] A. Fogel et al., “A general approach to network configuration analysis,” in *Proc. USENIX NSDI*, 2015.

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